Analyzing the use of quantitative geospatial analysis methods for measuring passenger rail demand and the value of connectivity

Brady Shields

College of Earth, Ocean, and Atmospheric Sciences

Oregon State University

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Introduction

The passenger rail transportation market in the United States has undergone major changes, including a successful post-pandemic demand rebound and a successful expansion and enhancement of services (Congressional Research Service, 2025). However, *Amtrak*, the nation's primary passenger rail service, is subsidized by the federal government, which can involve a different business management model than the private sector. Funding and authorization depend on Congress. However, there is a concept in transportation economics known as "induced demand." In relation to highway traffic, this concept is applied when measuring changes in traffic demand because of highway expansion. Changes in vehicle traffic and highway demand are an example of demand for transportation being elastic and not fixed (Lee, n.d.). From FY2019 to FY2024, Congress appropriated a total of roughly $30 billion for *Amtrak* (Congressional Research Service, n.d.). In contrast, from FY2019 to FY2024, Congress appropriated a total of roughly $315 billion for the *Federal Highway Administration* (Federal Highway Administration, n.d.).

In addition to induced demand, the *Congressional Research Service* has also acknowledged the value of connectivity for passenger rail (Congressional Research Service, 2025). Other studies have exhibited that new rail transportation can induce demand and expand entirely new markets, such as the increased real estate development and other land use changes indicating induced demand (Du, et. al., 2023). Despite this evidence, Congress seems unconvinced of the value of connectivity in *Amtrak* expansion, and congressional budget appropriations for *Amtrak* remains at a fraction of the value of budget appropriations for the *Federal Highway Administration*.

This project was aimed at exploring and analyzing ridership data for some of *Amtrak*'s recently expanded services. By exploring and analyzing the ridership data and comparing it to nearby

established services, we can examine the applicability of quantitative geospatial analysis methods to support the concepts of induced demand and connectivity value. This project's analysis was focused on the stations that are part of the newly expanded *Mardi Gras Service* from New Orleans, Louisiana to Mobile, Alabama. This new service has demonstrated success, and it has outperformed initial expectations (Southern Rail Commission, 2025). As such, ridership data analysis was limited to three states: Louisiana, Mississippi, and Alabama.

Figure 1: Exploratory map

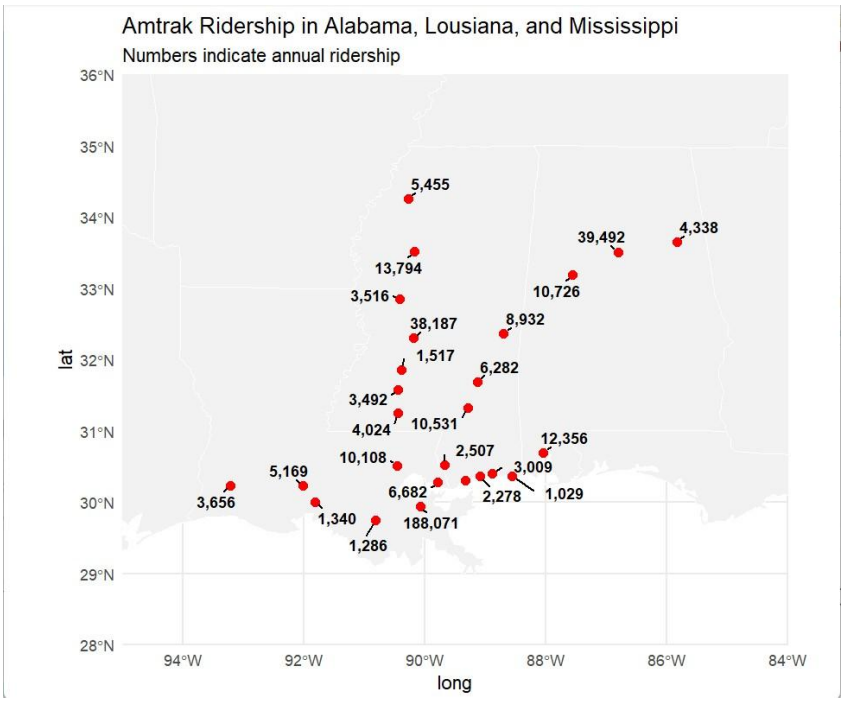

Fig. 1: This map was produced using *R* and *RStudio*, and it shows the distribution and annual ridership for Amtrak stations in Louisiana, Mississippi, and Alabama.

Methods

I used two primary datasets. The first one was annual *Amtrak* ridership data by station for 2025, which I acquired from the *Bureau of Transportation Statistics.* This will be referred to as "Dataset 1." The second one was census data for a variety of counties relevant to the study, which I acquired from the *US Census Bureau*. This will be referred to as "Dataset 2."

Dataset 1 initially included all of the annual ridership data for every station in the entire United States. I downloaded *Dataset 1* from the website of the *Bureau of Transportation Statistics*. The data was downloaded into a Comma-separated values (.csv) file format. I manually copied the data for the three states of interest: Louisiana, Mississippi, and Alabama. Then, I manually pasted the data into a new .csv file in Microsoft Excel that was limited to the three states of interest. The file included columns for latitude (lat), longitude (lon), state (state), station name (station), and annual ridership (ridership) for each station. *Lat, lon,* and *ridership* were quantitative variables, with *lat* and *lon* being quantitative spatial variables. *State* and *station* were categorical variables.

Dataset 2 was a combination of datasets downloaded from the comparison table tool on the *Quick Facts* section of the website of the *US Census Bureau*. Due to limitations on size of comparison tables, I created and downloaded 4 separate tables. These tables were for the following categories: Counties of interest in Alabama, counties of interest in Mississippi, northernmost counties of interest in Louisiana, and southernmost counties of interest in Louisiana. Each table was downloaded as a .csv file, and I copied total population for each county. Then, I pasted the total population numbers for each county in a new .csv file in *Microsoft Excel*. The new .csv file for *Dataset 2* now had data from *Dataset 1* for ridership as well. *Dataset 2* had 5 columns. These included columns for the state of which the county is a part (state), the county's name (county), total annual ridership for all *Amtrak* stations within the

county (ridership), total population of the county (population), and normalized ridership data for the county, measured in total annual ridership per 10,000 people (ride_10k). *State* and *county* are categorical variables, while *ridership, population,* and *ride_10k* are quantitative variables.

Dataset 1 and *Dataset 2* are my primary datasets that I downloaded from their primary sources, and I managed, filtered, and selected the datasets to create relevant and smaller datasets for my study area. In addition to the primary datasets, I also used the internal datasets within *R* and *RStudio* for state and county basemaps of the United States, which allowed for the joining of the basemap .csv files for *Dataset 1* and *Dataset 2*.

Figure 2: Dataset 1

	A	B	C	D	E	F
1	lat	lon	state	station	ridership	
2	33.649	-85.832	Alabama	Anniston,	4338	
3	33.5124	-86.807	Alabama	Birmingh:	39492	
4	30.6903	-88.038	Alabama	Mobile, Al	12356	
5	33.1923	-87.56	Alabama	Tuscaloo	10726	
6	30.5117	-90.464	Louisiana	Hammon	10108	
7	30.2273	-92.015	Louisiana	Lafayette,	5169	
8	30.2381	-93.215	Louisiana	Lake Cha	3656	
9	30.0085	-91.824	Louisiana	New Iberi	1340	
10	29.9468	-90.08	Louisiana	New Orle	188071	
11	29.7469	-90.813	Louisiana	Schriever	1286	
12	30.2782	-89.782	Louisiana	Slidell, Lc	6682	
13	30.3089	-89.334	Mississip	Bay St. Lc	2310	
14	30.3991	-88.891	Mississip	Biloxi, Mis	3009	
15	31.5788	-90.442	Mississip	Brookhav	3492	
16	33.5177	-90.176	Mississip	Greenwo	13794	
17	30.3692	-89.095	Mississip	Gulfport, I	2278	
18	31.3267	-89.286	Mississip	Hattiesbu	10531	
19	31.8629	-90.394	Mississip	Hazlehurs	1517	
20	32.3006	-90.191	Mississip	Jackson,	38187	
21	31.6924	-89.128	Mississip	Laurel, Mi	6282	
22	34.2582	-90.275	Mississip	Marks, Mi	5455	
23	31.2493	-90.453	Mississip	McComb,	4024	
24	32.3638	-88.696	Mississip	Meridian,	8932	
25	30.3678	-88.559	Mississip	Pascagou	1029	
26	30.5247	-89.68	Mississip	Picayune,	2507	
27	32.8481	-90.415	Mississip	Yazoo Cit	3516	

Fig. 2: Dataset 1 includes ridership data and spatial information about each station of interest. Figure is a screenshot of the .csv file in *Microsoft Excel*.

Figure 3: Dataset 2

	A	B	C	D	E	F
1	state	county	ridership	population	ride_10k	
2	Alabama	Calhoun	4338	116,427	372.594	
3	Alabama	Jefferson	39492	664,744	594.0934	
4	Alabama	Mobile	12356	412,339	299.6564	
5	Alabama	Tuscaloos	10726	241,212	444.6711	
6	Louisiana	Tangipaho	10108	139,823	722.914	
7	Louisiana	Lafayette	5169	254,241	203.311	
8	Louisiana	Calcasieu	3656	206,861	176.737	
9	Louisiana	Iberia	1340	67,540	198.4009	
10	Louisiana	Orleans	188071	362,701	5185.29	
11	Louisiana	Terrebonn	1286	103,864	123.8158	
12	Louisiana	St. Tamma	6682	277,615	240.693	
13	Mississipp	Hancock	2310	46,492	496.8597	
14	Mississipp	Harrison	5287	213,730	247.3682	
15	Mississipp	Jackson	1029	147,002	69.99905	
16						

Fig. 3: Dataset 2 includes rows for each county of interest, with columns for county name and state name, plus population of each county, total annual ridership of all *Amtrak* stations in each county, and normalized ridership data of ridership per 10,000 people. Figure is a screenshot of the .csv file in *Microsoft Excel*.

Initially, I made a choropleth map in *R* and *RStudio* with my joined datasets for ridership per 10,000 people by county. Then, I used the Local Moran's I test on the choropleth map to analyze hotspots and clusters of values.

Results

The choropleth map of ridership per 10,000 people in *Figure 4* shows slightly higher than average ridership per 10,000 people in some of the counties with stations along the *Mardi Gras Service* route. However, there is one outlier for the entire dataset, Orleans Parish in Louisiana, which is where New Orleans is located. As expected, New Orleans skews the mean, and in examining the data without New Orleans, the highest value is less than 600 riders per 10,000 people. In that case, only one county along the route is significantly low for normalized ridership per 10,000 people, and the rest are either around the mean or not significantly lower, or even significantly higher than the mean.

Figure 4: Choropleth map

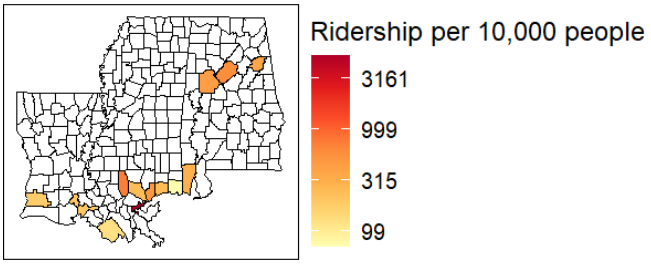

Fig. 4: Choropleth map made in *R* and *RStudio* that shows normalized ridership data by county, measured in ridership per 10,000 people.

In *Figure 5*, the map displays the results of the Local Moran's I hotspot test for significant hotspots of ridership per 10,000 people for each county. As expected, Orleans Parish, being the major outlier, is the only county considered a significant hotspot, with no other counties fitting such criteria.

Figure 5: Local Moran's I hotspot test

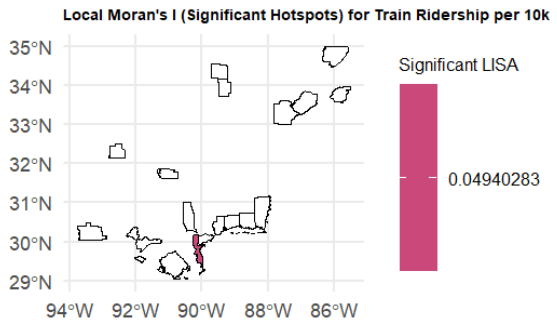

Fig. 5: Local Moran's I hotspot test for significant values for each county

After performing the hotspot test, I performed a cluster test with Local Moran's I. In *Figure 6*, the map shows clustering of high-high values where counties that are considered "clusters" are surrounded by high values. As expected, Orleans Parish, being an outlier, is the only county that is considered a "high-high" cluster and the rest are insignificant.

Figure 6: Local Moran's I cluster test

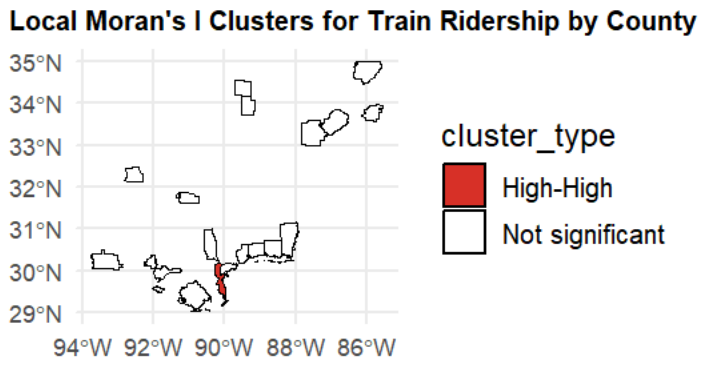

Fig. 6: Map that shows clustering of significant values by county.

Since Orleans Parish has been a consistent outlier for the choropleth map, the hotspot test, and the cluster test, I decided to run the cluster test but exclude Orleans Parish. After running the Local Moran's I cluster test but excluding New Orleans, I produced the plot in *Figure 7*. This, however, produced a result that involved no significant clusters.

Figure 7: Local Moran's I cluster test, excluding Orleans

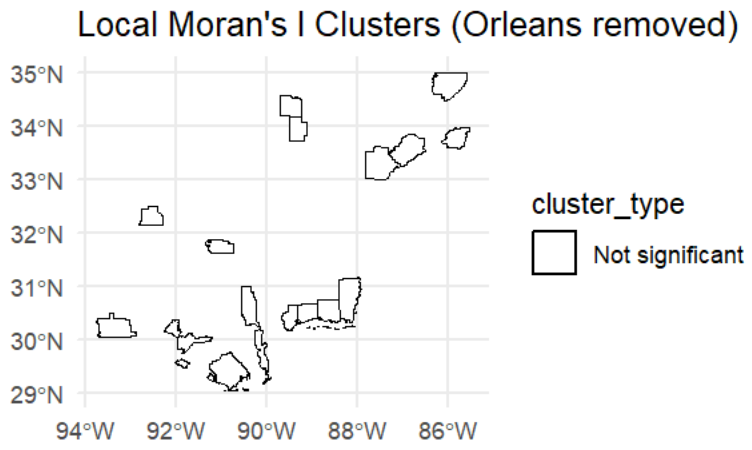

Fig. 7: A map showing the results of a Local Moran's I cluster test after removing Orleans Parish.

Discussion

By finding ridership and population data through primary sources with *Dataset 1* and *Dataset 2*, I was able to easily integrate the basemaps in *R* and *RStudio* for states and counties by joining the tables. In doing so, I created a choropleth map with normalized data for ridership per 10,000 people by county. Orleans was an outlier, and this led me to perform Local Moran's I hotspot and cluster tests. In both tests, when every county of interest was included, Orleans was the only significant hotspot and cluster. Therefore, I ran the cluster test while removing Orleans from the dataset. However, none of the counties were significant clusters in a test that removed Orleans.

These results from the Local Moran's I tests suggest that there is not significantly high ridership in the counties along the *Mardi Gras Service* route, but there is also not significantly low ridership along the same route. This can be interpreted to confirm the theory of induced demand, where new demand is created by the expansion of rail access. This is an example of demand elasticity. This confirms the previous public statements made by public agencies, where such agencies emphasized the service performing better with ridership than initially expected. This could be a result of initially assuming that demand for passenger rail is inelastic rather than elastic. This could be a reason to modify quantitative analysis modeling for estimating passenger rail demand to factor in the concept of induced demand and demand elasticity. Such a modification for quantitative analysis modeling would change how governments, policymakers, transportation economists, and railroad planners estimate demand and make decisions regarding rail expansion.

However, this method does have its limitations. As previously stated, while there were no significantly low clusters, there were also not any significantly high clusters. This suggests that the Local Moran's I test has many limitations with data that is more discrete and spatially separated.

If this project had been afforded infinite resources and time, more quantitative spatial analysis techniques could have been utilized to find one that might work better than the Local Moran's I tests.

By further analyzing spatial data regarding train ridership and exploring different quantitative spatial analysis techniques, we can explore and overcome the limitations of our current models that are used in our analysis to predict passenger rail demand. This will allow policymakers,

transportation economists, governments, and railroad planners to make better-informed decisions that factor in different variables.

References

Bureau of Transportation Statistics. (n.d.). Amtrak ridership. U.S. Department of Transportation. https://www.bts.gov/browse-statistical-products-and-data/state-transportation-statistics/amtrak-ridership

Congressional Research Service. (2025). Intercity passenger rail: Federal policy and programs (CRS Report No. R47260). Library of Congress. https://www.congress.gov/crs_external_products/R/HTML/R47260.web.html

Du, Q., Huang, Y., Zhou, Y., Guo, X., & Bai, L. (2023). Impacts of a new urban rail transit line and its interactions with land use on the ridership of existing stations. Cities, 141, 104506. https://doi.org/10.1016/j.cities.2023.104506

Federal Highway Administration. (n.d.). Funding and reports. U.S. Department of Transportation. https://highways.dot.gov/cfo/funding-and-reports

Lee, D. B., Jr. (n.d.). Induced traffic and induced demand [PDF]. National Association of City Transportation Officials. https://nacto.org/wp-content/uploads/induced_traffic_and_induced_demand_lee.pdf

Southern Rail Commission. (2025). Amtrak Mardi Gras service marks exceptional first-year performance [Press release]. https://www.southernrailcommission.org/press-releases/2025/12/5/amtrak-mardi-gras-service-marks-exceptional-first-year-performance

U.S. Census Bureau. (n.d.). QuickFacts. U.S. Department of Commerce. https://www.census.gov/quickfacts/fact/table/

Appendix

The following code is what I used in R, and it shows the whole process, including many errors and corrections.

```
##### Amtrak Ridership ####


#-------#

#Author: Brady Shields#

#Purpose: Explore Amtrak Ridership Data - and other data#

#Date: March 2026#


#------------------------#


# Load packages

library(sf)             # classes and functions for vector data

library(tmap)           # classes and functions for thematic mapping


##EXPLORATORY DATA


#####################################

# EXAMINE AND PREPARE THE DATA

# read in data from a CSV file
```

```
Full_data <-
read.csv("C:/Assignments/GIS/W26/Geog361/Project_Geog361/Project/Data/map_Full_Data_
data.csv")

names(Full_data)

table(Full_data)

write.csv(Full_data)

#LOAD NEW PACKAGES

library(sf)
library(tidyverse)
library(ggplot2)
library(maps)
library(ggrepel)

#CLEAN DATA

stations <-
read_csv("C:/Assignments/GIS/W26/Geog361/Project_Geog361/Project/Data/map_Full_Data
_data.csv") %>%
```

```
  select(Latitude, Longitude, State, Station, Value) %>%
  rename(
    lat = Latitude,
    lon = Longitude,
    ridership = Value,
    station = Station,
    state = State
  )

#CHECK DATA

glimpse(stations)

#CONVERT TO SPATIAL OBJECT

stations_sf <- st_as_sf(
  stations,
  coords = c("lon", "lat"),
  crs = 4326
)

#CREATE LABEL TEXT
```

```
stations_sf <- stations_sf %>%

  mutate(label = paste0(station, "\n", format(ridership, big.mark=",")))


#ADD BASEMAP


us_map <- map_data("state")


#NEW LABEL TEXT- AMENDED!


stations_sf <- stations_sf %>%

  mutate(label = format(ridership, big.mark = ","))


#CREATE CONUS EXTENT


xmin <- -125

xmax <- -66

ymin <- 24

ymax <- 50



#MAP WITH NUMERIC LABELS


ggplot() +
```

```
geom_polygon(data = us_map,
             aes(x = long, y = lat, group = group),
             fill = "gray95",
             color = "white",
             linewidth = 0.2) +

# station points
geom_sf(data = stations_sf,
        color = "red",
        size = 2.2) +

# ridership numbers
geom_text_repel(
  data = stations_sf,
  aes(geometry = geometry, label = label),
  stat = "sf_coordinates",
  size = 3.2,
  fontface = "bold",
  box.padding = 0.3,
  point.padding = 0.25,
  min.segment.length = 0
) +
```

```
  coord_sf(xlim = c(xmin, xmax), ylim = c(ymin, ymax), expand = FALSE) +


  theme_minimal() +
  labs(
    title = "U.S. Train Station Ridership",
    subtitle = "Numbers indicate annual ridership"
  )


#CREATE WEST COAST EXTENT


xmin <- -125
xmax <- -115
ymin <- 32
ymax <- 50


#MAP WITH NUMERIC LABELS


ggplot() +
  geom_polygon(data = us_map,
               aes(x = long, y = lat, group = group),
               fill = "gray95",
               color = "white",
               linewidth = 0.2) +
```

```
# station points
geom_sf(data = stations_sf,
        color = "red",
        size = 2.2) +

# ridership numbers
geom_text_repel(
  data = stations_sf,
  aes(geometry = geometry, label = label),
  stat = "sf_coordinates",
  size = 3.2,
  fontface = "bold",
  box.padding = 0.3,
  point.padding = 0.25,
  min.segment.length = 0
) +

coord_sf(xlim = c(xmin, xmax), ylim = c(ymin, ymax), expand = FALSE) +

theme_minimal() +
labs(
  title = "U.S. Train Station Ridership",
```

```
    subtitle = "Numbers indicate annual ridership"
  )

#CREATE NORTHWEST-MIDWEST N COR EXTENT

xmin <- -125
xmax <- -85
ymin <- 42
ymax <- 50

#MAP WITH NUMERIC LABELS

ggplot() +
  geom_polygon(data = us_map,
               aes(x = long, y = lat, group = group),
               fill = "gray95",
               color = "white",
               linewidth = 0.2) +

  # station points
  geom_sf(data = stations_sf,
          color = "red",
          size = 2.2) +
```

```
  # ridership numbers
  geom_text_repel(
    data = stations_sf,
    aes(geometry = geometry, label = label),
    stat = "sf_coordinates",
    size = 3.2,
    fontface = "bold",
    box.padding = 0.3,
    point.padding = 0.25,
    min.segment.length = 0
  ) +

  coord_sf(xlim = c(xmin, xmax), ylim = c(ymin, ymax), expand = FALSE) +

  theme_minimal() +
  labs(
    title = "U.S. Train Station Ridership",
    subtitle = "Numbers indicate annual ridership"
  )

#CREATE CENTRAL-EAST EXTENT
```

```
xmin <- -100

xmax <- -78

ymin <- 24

ymax <- 50


#MAP WITH NUMERIC LABELS


ggplot() +

  geom_polygon(data = us_map,

               aes(x = long, y = lat, group = group),

               fill ="gray95",

               color ="white",

               linewidth = 0.2) +


  # station points

  geom_sf(data = stations_sf,

          color = "red",

          size = 2.2) +


  # ridership numbers

  geom_text_repel(

    data = stations_sf,

    aes(geometry = geometry, label = label),
```

```
    stat = "sf_coordinates",
    size = 3.2,
    fontface = "bold",
    box.padding = 0.3,
    point.padding = 0.25,
    min.segment.length = 0
  ) +

  coord_sf(xlim = c(xmin, xmax), ylim = c(ymin, ymax), expand = FALSE) +

  theme_minimal() +
  labs(
    title = "U.S. Train Station Ridership",
    subtitle = "Numbers indicate annual ridership"
  )

#CREATE NE EXTENT

xmin <- -90
xmax <- -68
ymin <- 35
ymax <- 50
```

```
#MAP WITH NUMERIC LABELS


ggplot() +
  geom_polygon(data = us_map,
               aes(x = long, y = lat, group = group),
               fill ="gray95",
               color ="white",
               linewidth = 0.2) +


  # station points
  geom_sf(data = stations_sf,
          color = "red",
          size = 2.2) +


  # ridership numbers
  geom_text_repel(
    data = stations_sf,
    aes(geometry = geometry, label = label),
    stat = "sf_coordinates",
    size = 3.2,
    fontface = "bold",
    box.padding = 0.3,
    point.padding = 0.25,
```

```
    min.segment.length = 0
  ) +

  coord_sf(xlim = c(xmin, xmax), ylim = c(ymin, ymax), expand = FALSE) +

  theme_minimal() +
  labs(
    title = "U.S. Train Station Ridership",
    subtitle = "Numbers indicate annual ridership"
  )

#CREATE SW EXTENT

xmin <- -125
xmax <- -85
ymin <- 24
ymax <- 45

#MAP WITH NUMERIC LABELS

ggplot() +
  geom_polygon(data = us_map,
               aes(x = long, y = lat, group = group),
```

```
            fill ="gray95",
            color ="white",
            linewidth = 0.2) +

# station points
geom_sf(data = stations_sf,
        color = "red",
        size = 2.2) +

# ridership numbers
geom_text_repel(
  data = stations_sf,
  aes(geometry = geometry, label = label),
  stat = "sf_coordinates",
  size = 3.2,
  fontface = "bold",
  box.padding = 0.3,
  point.padding = 0.25,
  min.segment.length = 0
) +

coord_sf(xlim = c(xmin, xmax), ylim = c(ymin, ymax), expand = FALSE) +
```

```
  theme_minimal() +
  labs(
    title = "U.S. Train Station Ridership",
    subtitle = "Numbers indicate annual ridership"
  )



#MARDI GRAS SERVICE


#FILTER STATES
mardigras_states <- c("Alabama", "Mississippi", "Louisiana")


#CREATE EXTENT


xmin <- -90.3
xmax <- -87
ymin <- 29.8
ymax <- 31



stations_mardigras <- stations_sf %>%
  filter(state %in% mardigras_states)
```

```
#CROP BASEMAP


state_lookup_mardigras <- tibble(
  abb = state.abb,
  region = tolower(state.name)
)


selected_mardigras <- state_lookup_mardigras %>%
  filter(abb %in% mardigras_states) %>%
  pull(region)


mardigras_region_map <- us_map %>%
  filter(region %in% selected_mardigras)



#MARDI GRAS MAP


ggplot() +
  geom_polygon(data = mardigras_region_map,
               aes(x = long, y = lat, group = group),
               fill = "gray95",
               color = "white",
               linewidth = 0.2) +
```

```
  geom_sf(data = stations_mardigras,
          color = "red",
          size = 2.5) +

  geom_text_repel(
    data = stations_mardigras,
    aes(geometry = geometry, label = label),
    stat = "sf_coordinates",
    size = 3.4,
    fontface = "bold",
    force = 2,
    max.overlaps = Inf,
    min.segment.length = 0
  ) +

  coord_sf(xlim = c(xmin, xmax), ylim = c(ymin, ymax), expand = FALSE) +
  theme_minimal() +
  labs(
    title = "Train Station Ridership — Mardi Gras Service Map"
  )

#CITY OF NEW ORLEANS ROUTE
```

```
#FILTER STATES
cityofnola_states <- c("Illinois", "Kentucky", "Tennessee", "Mississippi", "Louisiana")

stations_cityofnola <- stations_sf %>%
  filter(state %in% cityofnola_states)

#CREATE EXTENT

xmin <- -91
xmax <- -86
ymin <- 29.8
ymax <- 42.25

#CITY OF NEW ORLEANS ROUTE MAP

ggplot() +
  geom_polygon(data = cityofnola_map,
               aes(x = long, y = lat, group = group),
               fill = "gray95",
               color = "white",
               linewidth = 0.2) +
```

```
  geom_sf(data = stations_cityofnola,
          color = "red",
          size = 2.5) +

  geom_text_repel(
    data = stations_cityofnola,
    aes(geometry = geometry, label = label),
    stat = "sf_coordinates",
    size = 3.4,
    fontface = "bold",
    force = 2,
    max.overlaps = Inf,
    min.segment.length = 0
  ) +

  coord_sf(xlim = c(xmin, xmax), ylim = c(ymin, ymax), expand = FALSE) +
  theme_minimal() +
  labs(
    title = "Train Station Ridership — City of New Orleans Route Map"
  )

#CROP BASEMAP
```

```
state_lookup_cityofnola <- tibble(
  abb = state.abb,
  region = tolower(state.name)
)


selected_cityofnola <- state_lookup_cityofnola %>%
  filter(abb %in% mardigras_states) %>%
  pull(region)


cityofnola_map <- us_map %>%
  filter(region %in% selected_cityofnola)



#CITY OF NEW ORLEANS ROUTE MAP


ggplot() +
  geom_polygon(data = cityofnola_map,
               aes(x = long, y = lat, group = group),
               fill = "gray95",
               color = "white",
               linewidth = 0.2) +


  geom_sf(data = stations_cityofnola,
```

```
        color = "red",
        size = 2.5) +

  geom_text_repel(
    data = stations_cityofnola,
    aes(geometry = geometry, label = label),
    stat = "sf_coordinates",
    size = 3.4,
    fontface = "bold",
    force = 2,
    max.overlaps = Inf,
    min.segment.length = 0
  ) +

  coord_sf(xlim = c(xmin, xmax), ylim = c(ymin, ymax), expand = FALSE) +
  theme_minimal() +
  labs(
    title = "Train Station Ridership — City of New Orleans Route Map"
  )
#
#######################################
#
#NEW STUDY AREA
```

```
################################
#
#LIMIT STATIONS TO THREE STATES RELEVANT TO AREA
stations_3states <- stations %>%
  filter(state %in% c("Alabama", "Louisiana", "Mississippi"))


#CREATE SHAPEFILE FOR LIMITED EXTENT
stations3_sf <- st_as_sf(
  stations_3states,
  coords = c("lon", "lat"),
  crs = 4326
)


#NEW LABEL TEXT
stations3_sf <- stations3_sf %>%
  mutate(label = format(ridership, big.mark = ","))


#CREATE EXTENT FOR THREE STATES


xmin <- -95
xmax <- -84
ymin <- 28
ymax <- 36
```

```
#MAP WITH NUMERIC LABELS FOR THREE STATES
library(ggrepel)


ggplot() +
  geom_polygon(data = us_map,
               aes(x = long, y = lat, group = group),
               fill = "gray95",
               color = "white",
               linewidth = 0.2) +


  # station points
  geom_sf(data = stations3_sf,
          color = "red",
          size = 2.2) +


  # ridership numbers
  geom_text_repel(
    data = stations3_sf,
    aes(geometry = geometry, label = label),
    stat = "sf_coordinates",
    size = 3,
    fontface = "bold",
```

```
    box.padding = 0.3,
    point.padding = 0.25,
    min.segment.length = 0
  ) +

  coord_sf(xlim = c(xmin, xmax), ylim = c(ymin, ymax), expand = FALSE) +

  theme_minimal() +
  labs(
    title = "Amtrak Ridership in Alabama, Lousiana, and Mississippi",
    subtitle = "Numbers indicate annual ridership"
  ) +
  theme(plot.title = element_text(size = 9),
        plot.subtitle = element_text(size = 7)
        )


#
###################
#
#MAKE MAP OF COUNTY BOUNDARIES
states3_counties <- subset(
  counties,
```

```
  region %in% c("alabama", "louisiana", "mississippi")
)
#
#PLOT THE COUNTY MAP
ggplot() +
  geom_polygon(
    data = states3_counties,
    aes(x = long, y = lat, group = group),
    fill = "white",
    color = "black",
    linewidth = 0.2
  ) +
  coord_fixed(1.3) +
  theme_minimal()
#
#Get CSV file
write_csv(
  stations_3states,
  "C:/Assignments/GIS/W26/Geog361/Project_Geog361/Project/Data/stations_3states.csv"
)
#Additional step
stations_3states_sf <- stations_3states %>%
st_as_sf(coords = c("lon", "lat"), crs = 4326)
```

```
#

#

###################

#

#CHANGE BASEMAP

#

ggplot() +

  geom_polygon(

    data = states3_counties,

    aes(x = long, y = lat, group = group),

    fill = "gray95",

    color = "white",

    linewidth = 0.2

  ) +

  geom_sf(

    data = stations_3states_sf,

    color = "red",

    size = 2.5

  ) +

  coord_sf(expand = FALSE) +

  theme_minimal() +

  labs(

    title = "Amtrak Stations with ridership",
```

```
    subtitle = "Alabama, Louisiana, Mississippi"
  ) +
  geom_text_repel(
    data = stations_3states_sf,
    aes(geometry = geometry, label = label),
    stat = "sf_coordinates",
    size = 3,
    min.segment.length = 0
  )
#
#Check for errors
stations_3states_sf <- stations_3states_sf %>%
  mutate(label = format(ridership, big.mark = ","))
#
#


###################
#
#POPULATION OF EACH LOCALITY
write.csv(Countypopulations)
read.csv(Countypopulations)


#       "State"        "County.Parish"    "Population"
```

```
# "1"   "Alabama"      "Calhoun"          "116,427"

# "2"   "Alabama"      "Jefferson"        "664,744"

# "3"   "Alabama"      "Tuscaloosa"       "241,212"

# "4"   "Alabama"      "Mobile"           "412,339"

# "5"   "Mississippi"  "Jackson"          "147,002"

# "6"   "Mississippi"  "Harrison"         "213,730"

# "7"   "Mississippi"  "Hancock"          "46,492"

# "8"   "Louisiana"    "Tangipahoa"       "139,823"

# "9"   "Louisiana"    "St Tammany"       "277,615"

# "10"  "Louisiana"    "Calcasieu"        "206,861"

# "11"  "Louisiana"    "Lafayette"        "254,241"

# "12"  "Louisiana"    "Iberia"           "67,540"

# "13"  "Louisiana"    "Orleans"          "362,701"

# "14"  "Louisiana"    "Terrebonne"       "103,864"


write.csv(stations_3states)


#Ridership by county data

write.csv(ridership_county)


#Join data

#read CSV
```

```
read.csv("C:/Assignments/GIS/W26/Geog361/Project_Geog361/Project/Data/ridership_county2.csv")

library(sf)

counties <- st_read("states3_counties")

ls(states3_counties)

library(dplyr)

counties_joined <- left_join(states3_counties,
                             ridership_county2,
                             by = c("region" = "state",
                                    "subregion" = "county"))

ridership_county2 <-
read.csv("C:/Assignments/GIS/W26/Geog361/Project_Geog361/Project/Data/ridership_county2.csv", header = TRUE)

names(ridership_county2)

ridership_county2$state <- tolower(ridership_county2$state)
```

```
ridership_county2$county <- tolower(ridership_county2$county)


library(dplyr)


counties_joined <- left_join(states3_counties,
                             ridership_county2,
                             by = c("region"="state",
                                    "subregion"="county"))



summary(counties_joined$ride_10k)


head(ridership_county2)


ridership_county2$population <- as.numeric(gsub(",", "", ridership_county2$population))


str(ridership_county2)


library(dplyr)


counties_joined <- left_join(states3_counties,
                             ridership_county2,
                             by = c("region" = "state",
```

```
                                "subregion" = "county"))


summary(counties_joined$ride_10k)


library(ggplot2)


ggplot(counties_joined,
       aes(x = long, y = lat, group = group, fill = ride_10k)) +
  geom_polygon(color = "black") +
  coord_fixed(1.3) +
  scale_fill_viridis_c(name = "Ridership per 10,000") +
  theme_void()


unique(states3_counties$subregion)


ridership_county2$county[ridership_county2$county == "st tammany"] <- "st tammany"


unique(ridership_county2$county)


ridership_county2$county[ridership_county2$county == "st tammany"] <- "st. tammany"


library(dplyr)
```

```
counties_joined <- left_join(states3_counties,
                        ridership_county2,
                        by = c("region"="state",
                              "subregion"="county"))


counties_joined[counties_joined$subregion == "st. tammany", ]


unique(states3_counties$subregion[states3_counties$region == "louisiana"])


unique(ridership_county2$county)


ridership_county2$county[ridership_county2$county == "st. tammany"] <- "st tammany"


library(dplyr)


counties_joined <- left_join(states3_counties,
                        ridership_county2,
                        by = c("region" = "state",
                              "subregion" = "county"))


counties_joined[counties_joined$subregion == "st tammany", ]


library(ggplot2)
```

```
ggplot(counties_joined,
       aes(long, lat, group = group, fill = ride_10k)) +
  geom_polygon(color = "black") +
  coord_fixed(1.3) +
  scale_fill_viridis_c(name = "Ridership per 10,000 people") +
  theme_void()

counties_filtered <- states3_counties %>%
  dplyr::filter(subregion %in% ridership_county2$county &
                region %in% ridership_county2$state)

counties_joined <- dplyr::left_join(counties_filtered,
                                    ridership_county2,
                                    by = c("region"="state",
                                           "subregion"="county"))

library(ggplot2)

ggplot(counties_joined,
       aes(x = long, y = lat, group = group, fill = ride_10k)) +
  geom_polygon(color = "black") +
  coord_fixed(1.3) +
```

```
  scale_fill_viridis_c(name = "Ridership per 10,000 people",
                       na.value = "white") +
  theme_void()


ggplot(counties_joined,
       aes(x = long, y = lat, group = group, fill = ride_10k)) +
  geom_polygon(color = "black") +
  coord_fixed(1.3) +
  scale_fill_viridis_c(name = "Ridership per 10,000 people") +
  theme_void()


library(ggplot2)


ggplot(counties_joined,
       aes(long, lat, group = group, fill = ride_10k)) +
  geom_polygon(color = "black") +
  coord_fixed(1.3) +
  scale_fill_viridis_c(name = "Ridership per 10,000 people") +
  theme_void()


library(ggplot2)


ggplot(counties_joined,
```

```
      aes(x = long, y = lat, group = group, fill = ride_10k)) +
  geom_polygon(color = "black", size = 0.2) +
  coord_fixed(1.3) +
  scale_fill_viridis_c(name = "Ridership per 10,000",
                       na.value = "white") +
  theme_void() +
  theme(panel.background = element_rect(fill = "white"))


library(ggplot2)


counties <- map_data("county")


states3_counties <- subset(counties,
                           region %in% c("alabama","louisiana","mississippi"))


library(dplyr)


counties_joined <- left_join(states3_counties,
                             ridership_county2,
                             by = c("region"="state",
                                    "subregion"="county"))


ggplot(counties_joined,
```

```
      aes(long, lat, group = group, fill = ride_10k)) +
  geom_polygon(color="black", size=0.2) +
  coord_fixed(1.3) +
  scale_fill_viridis_c(name="Ridership per 10,000",
                       na.value="white") +
  theme_void()


counties_joined$ride_class <- cut(counties_joined$ride_10k,
                                  breaks = 5)




ggplot(counties_joined,
      aes(long, lat, group = group, fill = ride_class)) +
  geom_polygon(color = "black", size = 0.2) +
  coord_fixed(1.3) +
  scale_fill_brewer(palette = "YlOrRd",
                    na.value = "white",
                    name = "Ridership per 10,000 people") +
  theme_void()


counties_joined$ride_log <- log10(counties_joined$ride_10k + 1)


library(ggplot2)
```

```
ggplot(counties_joined,
       aes(long, lat, group = group, fill = ride_log)) +
  geom_polygon(color = "black", size = 0.2) +
  coord_fixed(1.3) +
  scale_fill_distiller(
    palette = "YlOrRd",
    na.value = "white",
    direction = 1,
    labels = function(x) round(10^x - 1, 0),
    name = "Ridership per 10,000 people"
  )+
  theme_void() +
  theme(
    legend.position = "right",
    legend.title = element_text(size = 12),
    legend.text = element_text(size = 10),
    panel.background = element_rect(fill = "white")
  )


#Global Moran's Test
library(spdep)
library(sf)
```

```
library(tigris)

library(sf)

library(dplyr)


# Get counties for the three states

al <- counties(state = "AL", cb = TRUE)

la <- counties(state = "LA", cb = TRUE)

ms <- counties(state = "MS", cb = TRUE)


# Combine into one sf object

counties_sf <- rbind(al, la, ms)


# Join ridership data

ridership_county2$state_fips <- tolower(ridership_county2$state)  # make sure to match

ridership_county2$county_name <- tolower(ridership_county2$county)


counties_sf <- counties_sf %>%

  mutate(NAME_lower = tolower(NAME)) %>%

  left_join(ridership_county2, by = c("NAME_lower" = "county_name"))


# Create neighbors and weights

library(spdep)
```

```
# Create neighbors list (queen contiguity)
nb <- poly2nb(counties_sf, queen = TRUE)

# Create spatial weights
lw <- nb2listw(nb, style = "W", zero.policy = TRUE)

# Keep only counties with ride_10k
counties_sf_moran <- counties_sf %>%
  filter(!is.na(ride_10k))

# Recreate neighbors for this subset
nb_sub <- poly2nb(counties_sf_moran, queen = TRUE)
lw_sub <- nb2listw(nb_sub, style = "W", zero.policy = TRUE)

# Run Global Moran's I
global_moran <- moran.test(counties_sf_moran$ride_10k, lw_sub, zero.policy = TRUE)
global_moran

library(spdep)

# Full neighbor list (all counties, not just 14)
nb_full <- poly2nb(counties_sf, queen = TRUE)
```

```
lw_full <- nb2listw(nb_full, style = "W", zero.policy = TRUE)


# Global Moran's I ignoring NA values

global_moran <- moran.test(counties_sf$ride_10k, lw_full, na.action = na.exclude, zero.policy

= TRUE)

global_moran


# ----------------------------

# Local Moran's I (LISA) for Ridership

# ----------------------------


library(sf)

library(dplyr)

library(spdep)

library(ggplot2)


# 1. Make sure your counties_sf has ride_10k and geometry

# Assuming you already joined your ridership data


# 2. Create neighbors using the full three-state polygons

nb_full <- poly2nb(counties_sf, queen = TRUE)

lw_full <- nb2listw(nb_full, style = "W", zero.policy = TRUE)
```

```
# 3. Calculate Local Moran's I

local <- localmoran(counties_sf$ride_10k, lw_full, zero.policy = TRUE)



# 4. Attach results to counties_sf

counties_sf$localI <- local[,1]       # Local Moran's I statistic

counties_sf$local_p <- local[,5]       # p-value

counties_sf$significant <- ifelse(counties_sf$local_p < 0.05, counties_sf$localI, NA)



# 5. Map significant hotspots

ggplot(counties_sf) +

  geom_sf(aes(fill = significant), color = "black") +

  scale_fill_viridis_c(option = "plasma", na.value = "white", name = "Significant LISA") +

  theme_minimal() +

  ggtitle("Local Moran's I (Significant Hotspots) for Train Ridership per 10k")

#####

#Failed due to data set issues

####

#New Attempt



library(sf)

library(dplyr)

library(spdep)

library(ggplot2)
```

```
# Step 1: Keep only counties with ride_10k
counties_sf_moran <- counties_sf %>%
  filter(!is.na(ride_10k))

# Step 2: Create neighbors for this subset
nb_sub <- poly2nb(counties_sf_moran, queen = TRUE)
lw_sub <- nb2listw(nb_sub, style = "W", zero.policy = TRUE)

# Step 3: Local Moran's I
local <- localmoran(counties_sf_moran$ride_10k, lw_sub, zero.policy = TRUE)
local

# Step 4: Attach results
counties_sf_moran$localI <- local[,1]
counties_sf_moran$local_p <- local[,5]
counties_sf_moran$significant <- ifelse(counties_sf_moran$local_p < 0.05,
                                        counties_sf_moran$localI, NA)

# Step 5: Map significant hotspots
ggplot(counties_sf_moran) +
  geom_sf(aes(fill = significant), color = "black") +
  scale_fill_viridis_c(option = "plasma", na.value = "white", name = "Significant LISA") +
```

```
  theme_minimal() +
  theme(
    plot.margin = margin(20, 20, 20, 20),  # top, right, bottom, left
    legend.position = "right",
    legend.title = element_text(size = 8),
    legend.text = element_text(size = 8),
    plot.title = element_text(size = 7, face = "bold")
  ) +
  ggtitle("Local Moran's I (Significant Hotspots) for Train Ridership per 10k")


#Only shows NOLA


# ----------------------------
# Local Moran Quadrant Map (High-High / Low-Low)
# ----------------------------


library(sf)
library(dplyr)
library(spdep)
library(ggplot2)


# Step 1: Filter only counties with ridership data
counties_sf_moran <- counties_sf %>%
```

```
  filter(!is.na(ride_10k))


# Step 2: Neighbors and weights

nb_sub <- poly2nb(counties_sf_moran, queen = TRUE)

lw_sub <- nb2listw(nb_sub, style = "W", zero.policy = TRUE)


# Step 3: Local Moran's I

local <- localmoran(counties_sf_moran$ride_10k, lw_sub, zero.policy = TRUE)


# Step 4: Add results

counties_sf_moran$localI <- local[,1]

counties_sf_moran$local_p <- local[,5]


# Step 5: Classify clusters (only significant p < 0.05)

mean_val <- mean(counties_sf_moran$ride_10k, na.rm = TRUE)


counties_sf_moran <- counties_sf_moran %>%

  mutate(cluster_type = case_when(

    local_p < 0.05 & ride_10k > mean_val & localI > 0 ~ "High-High",

    local_p < 0.05 & ride_10k < mean_val & localI > 0 ~ "Low-Low",

    local_p < 0.05 & ride_10k > mean_val & localI < 0 ~ "High-Low",

    local_p < 0.05 & ride_10k < mean_val & localI < 0 ~ "Low-High",

    TRUE ~ "Not significant"
```

```
  ))


# Step 6: Plot
ggplot(counties_sf_moran) +
  geom_sf(aes(fill = cluster_type), color = "black") +
  scale_fill_manual(values = c(
    "High-High" = "#d73027",  # red
    "Low-Low"   = "#4575b4",  # blue
    "High-Low"  = "#fc8d59",  # orange
    "Low-High"  = "#91bfdb",  # light blue
    "Not significant" = "white"
  )) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 10, face = "bold"),
    plot.title.position = "plot",   # <- makes title use the full plot space
    plot.margin = margin(20, 20, 20, 20),  # top, right, bottom, left,
    legend.position = "right",
    legend.title = element_text(size = 12),
    legend.text = element_text(size = 10)) +
  labs(title = "Local Moran's I Clusters for Train Ridership by County")
```

```
##########

#Without NOLA

########

counties_sf_no_orl <- counties_sf_moran %>%

  filter(NAME != "Orleans")


nb_sub_out <- poly2nb(counties_sf_no_orl, queen = TRUE)

lw_sub_out <- nb2listw(nb_sub_out, style = "W", zero.policy = TRUE)


local_out <- localmoran(counties_sf_no_orl$ride_10k, lw_sub_out, zero.policy = TRUE)


counties_sf_no_orl$localI <- local[,1]

counties_sf_no_orl$local_p <- local[,5]


counties_sf_no_orl$cluster_type <- case_when(

  local_p < 0.05 & ride_10k > mean(ride_10k) & localI > 0 ~ "High-High",

  local_p < 0.05 & ride_10k < mean(ride_10k) & localI > 0 ~ "Low-Low",

  local_p < 0.05 & ride_10k > mean(ride_10k) & localI < 0 ~ "High-Low",

  local_p < 0.05 & ride_10k < mean(ride_10k) & localI < 0 ~ "Low-High",

  TRUE ~ "Not significant"

)
```

```
####

#New attmept

##


# 1. Filter out Orleans

counties_sf_no_orl <- counties_sf %>% filter(NAME != "Orleans")


# 2. Rebuild neighbors and weights for this subset

nb_sub_out <- poly2nb(counties_sf_no_orl, queen = TRUE)

lw_sub_out <- nb2listw(nb_sub_out, style = "W", zero.policy = TRUE)


# 3. Run Local Moran on the filtered dataset

local_out <- localmoran(counties_sf_no_orl$ride_10k, lw_sub_out, zero.policy = TRUE)


# 4. Attach results

counties_sf_no_orl$localI <- local[,1]

counties_sf_no_orl$local_p <- local[,5]


# 5. Classify clusters

mean_val <- mean(counties_sf_no_orl$ride_10k, na.rm = TRUE)

counties_sf_no_orl$cluster_type <- case_when(

  local_p < 0.05 & ride_10k > mean_val & localI > 0 ~ "High-High",

  local_p < 0.05 & ride_10k < mean_val & localI > 0 ~ "Low-Low",
```

```
  local_p < 0.05 & ride_10k > mean_val & localI < 0 ~ "High-Low",
  local_p < 0.05 & ride_10k < mean_val & localI < 0 ~ "Low-High",
  TRUE ~ "Not significant"
)




####
#another attempt to filter data
##


library(spdep)
library(dplyr)


# 1. Keep only counties with data and remove Orleans
counties_sf_no_orl <- counties_sf %>%
  filter(!is.na(ride_10k) & NAME != "Orleans")


# 2. Rebuild neighbors and weights for this filtered subset
nb_sub_out <- poly2nb(counties_sf_no_orl, queen = TRUE)
lw_sub_out <- nb2listw(nb_sub_out, style = "W", zero.policy = TRUE)


# 3. Run Local Moran's I
local_out <- localmoran(counties_sf_no_orl$ride_10k, lw_sub_out, zero.policy = TRUE)
```

```
# 4. Attach results
counties_sf_no_orl$localI <- local_out[,1]
counties_sf_no_orl$local_p <- local_out[,5]

# 5. Classify clusters (high-high / low-low)
mean_val <- mean(counties_sf_no_orl$ride_10k)
counties_sf_no_orl$cluster_type <- case_when(
  local_p < 0.05 & ride_10k > mean_val & localI > 0 ~ "High-High",
  local_p < 0.05 & ride_10k < mean_val & localI > 0 ~ "Low-Low",
  local_p < 0.05 & ride_10k > mean_val & localI < 0 ~ "High-Low",
  local_p < 0.05 & ride_10k < mean_val & localI < 0 ~ "Low-High",
  TRUE ~ "Not significant"
)

####
#Local p failure attempt again
#######
library(sf)
library(dplyr)
library(spdep)
library(ggplot2)
```

```
# 1. Filter to keep only counties with ridership data, exclude Orleans

counties_sf_no_orl <- counties_sf %>%

  filter(!is.na(ride_10k) & NAME != "Orleans")


# 2. Build neighbors and weights

nb_sub_out <- poly2nb(counties_sf_no_orl, queen = TRUE)

lw_sub_out <- nb2listw(nb_sub_out, style = "W", zero.policy = TRUE)


# 3. Calculate Local Moran's I

local_out <- localmoran(counties_sf_no_orl$ride_10k, lw_sub_out, zero.policy = TRUE)


# 4. Attach results to the sf object

counties_sf_no_orl$localI <- local_out[,1]

counties_sf_no_orl$local_p <- local_out[,5]


# 5. Classify clusters

mean_val <- mean(counties_sf_no_orl$ride_10k)

counties_sf_no_orl <- counties_sf_no_orl %>%

  mutate(cluster_type = case_when(

    local_p < 0.05 & ride_10k > mean_val & localI > 0 ~ "High-High",

    local_p < 0.05 & ride_10k < mean_val & localI > 0 ~ "Low-Low",

    local_p < 0.05 & ride_10k > mean_val & localI < 0 ~ "High-Low",

    local_p < 0.05 & ride_10k < mean_val & localI < 0 ~ "Low-High",
```

```
    TRUE ~ "Not significant"
  ))

# 6. Plot clusters
ggplot(counties_sf_no_orl) +
  geom_sf(aes(fill = cluster_type), color = "black") +
  scale_fill_manual(values = c(
    "High-High" = "#d73027",
    "Low-Low"   = "#4575b4",
    "High-Low"  = "#fc8d59",
    "Low-High"  = "#91bfdb",
    "Not significant" = "white"
  )) +
  theme_minimal() +
  ggtitle("Local Moran's I Clusters (Orleans removed)")
```